

Report R2 — Unitary sweep and fragmentation scaling

Krylov sector analysis of quantum cellular automata (Tier 1a/1c)

QCA Sector Analysis project (Tier 1)

2026-07-12 (engine_version 1a.1)

Abstract

We sweep all sixteen unitary quantum elementary cellular automata over chain lengths $N = 6 \dots 18$, both boundary conventions, and classify each by the scaling of its Krylov-sector structure. Sector sizes are exact; the growth of the number of sectors and of the largest sector $D_{\max}$ is fit by BIC model selection ($\ln y = c[+\alpha \ln N][+\kappa N]$). Seven rules are effectively ergodic (a single sector spanning the space); the remainder fall into three fragmentation families — exponential (156/198, 108, 201), a domain-wall family with a linearly growing sector count but an extensive $D_{\max} = 2^{N-1}$ (60/102, `obc0` only), and 150/105 with central-binomial sectors. We highlight two boundary-sensitive phenomena and one broken symmetry.

1 Method

Each unit (rule, N , bc) is computed by the exact Tier-1a engine (Report R1) and cached as one JSON line in `results/`. Unitary sectors are the weakly connected components of the one-cycle transition graph (extracted by union-find; see R1). A rule is flagged *effectively ergodic at N* when one sector exceeds 90% of the 2^N states; the sweep then stops increasing N for that rule/bc. Growth classes are assigned to the series $y(N) \in \{\#\text{sectors}, D_{\max}\}$ by BIC selection among constant / power-law / exponential models, with the $\alpha \ln N$ prefactor always carried when reporting the exponential rate κ (so that $\sqrt{N}$ -type binomial prefactors do not bias κ).

2 Results

Table 1 lists every computed rule/bc with its sector count and $D_{\max}$ at the largest N reached and the fitted growth classes. Figures 1–3 show the scaling.

3 Boundary-sensitive fragmentation

Two rules change fragmentation class with the boundary:

- **Rules 60/102 (IIVV)/(IVIV):** under `obc0` the domain-wall charge fragments the space into $N+1$ sectors with an extensive dominant sector $D_{\max} = 2^{N-1}$; under `pbc` the periodic seam destroys the charge and the dynamics is ergodic — a single sector of size $2^N - 1$ plus the frozen vacuum $|0 \dots 0\rangle$ ($D_{\max}/2^N \rightarrow 1$).
- **Rules 150/105:** identical sector data under `obc0` (central binomials $\binom{N+1}{w}$), but they *differ* under `pbc`, where the periodic domain-wall counting is rule-specific.

W	tuple	bc	$N_{\max}$	#sec	$D_{\max}$	#sec growth	$D_{\max}$ growth (base)
150	IVVI	obc0	18	10	92378	polynomial	exponential (2.011)
60	IIVV	obc0	18	19	131072	polynomial	exponential (2.000)
102	IVIV	obc0	18	19	131072	polynomial	exponential (2.000)
105	VIIV	obc0	18	10	92378	polynomial	exponential (1.973)
201	VIII	obc0	18	17991	6765	exponential	exponential (1.618)
108	IIIV	obc0	18	31572	2584	exponential	exponential (1.617)
156	IIVI	obc0	18	6765	192	exponential	exponential (1.281)
198	IVII	obc0	18	6765	192	exponential	exponential (1.281)
204	IIII	obc0	18	262144	1	exponential	constant
51	VVVV	obc0 ^{†6}	–	–	–	ergodic	ergodic
54	IVVV	obc0 ^{†6}	–	–	–	ergodic	ergodic
57	VIIV	obc0 ^{†6}	–	–	–	ergodic	ergodic
99	VVIV	obc0 ^{†6}	–	–	–	ergodic	ergodic
147	VVVI	obc0 ^{†6}	–	–	–	ergodic	ergodic
153	VIVI	obc0 ^{†6}	–	–	–	ergodic	ergodic
195	VVII	obc0 ^{†6}	–	–	–	ergodic	ergodic
105	VIIV	pbc	18	9	97240	constant	exponential (1.998)
150	IVVI	pbc	18	12	87516	polynomial	exponential (1.888)
108	IIIV	pbc	18	24914	5778	exponential	exponential (1.619)
201	VIII	pbc	18	24914	5778	exponential	exponential (1.619)
156	IIVI	pbc	18	5779	144	exponential	exponential (1.293)
198	IVII	pbc	18	5779	144	exponential	exponential (1.293)
204	IIII	pbc	18	262144	1	exponential	constant
51	VVVV	pbc ^{†6}	–	–	–	ergodic	ergodic
54	IVVV	pbc ^{†6}	–	–	–	ergodic	ergodic
57	VIIV	pbc ^{†6}	–	–	–	ergodic	ergodic
60	IIVV	pbc ^{†6}	–	–	–	ergodic	ergodic
99	VVIV	pbc ^{†6}	–	–	–	ergodic	ergodic
102	IVIV	pbc ^{†6}	–	–	–	ergodic	ergodic
147	VVVI	pbc ^{†6}	–	–	–	ergodic	ergodic
153	VIVI	pbc ^{†6}	–	–	–	ergodic	ergodic
195	VVII	pbc ^{†6}	–	–	–	ergodic	ergodic

Table 1: Per-rule fragmentation scaling (unitary rules). [†] N marks the N at which a rule first becomes effectively ergodic for that boundary convention. Rules reduced by reflection are represented once where noted.

4 Broken inversion symmetry (201 vs. 108)

Rules 201 = (VIII) and 108 = (IIIV) are $0 \leftrightarrow 1$ spin-flip partners. Because the flip conjugates the Hadamard by X and $XHX \neq \pm H$, the symmetry is broken for the full dynamics. At the level of the sector partition it survives under **pbc** (bit-flip is a graph symmetry: $|XHX| = |H|$) but is broken under **obc0** by the vacuum-0 padding: there 201 has Fibonacci $D_{\max}$ and 108 shifted-Fibonacci $D_{\max}$ (matching **HSF** rules 1 and 8). See Report R1 for the exact statement.

5 Reproducibility

All numbers derive from `results/*.jsonl` (engine_version 1a.1); the table and figures are regenerated by `python -m qca.fragmentation.scaling.summary` and `...scaling.figure`. The sweep is idempotent and checkpointed (`checkpoints/manifest.jsonl`).

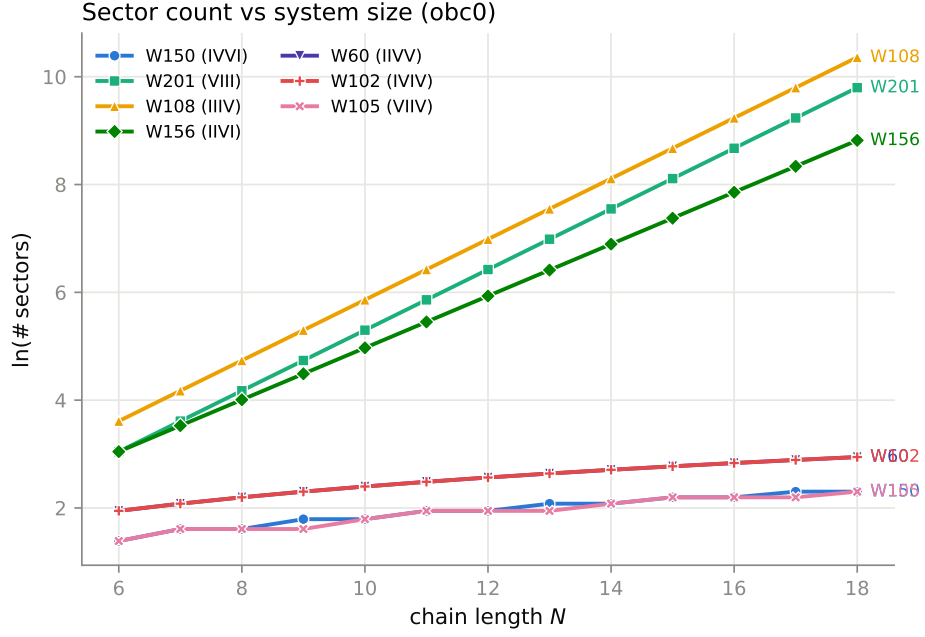


Figure 1: Number of Krylov sectors vs N (obc0), log scale. Exponential fragmentation (156, 108, 201) vs. linear/central-binomial sector counts (60/102, 150/105).

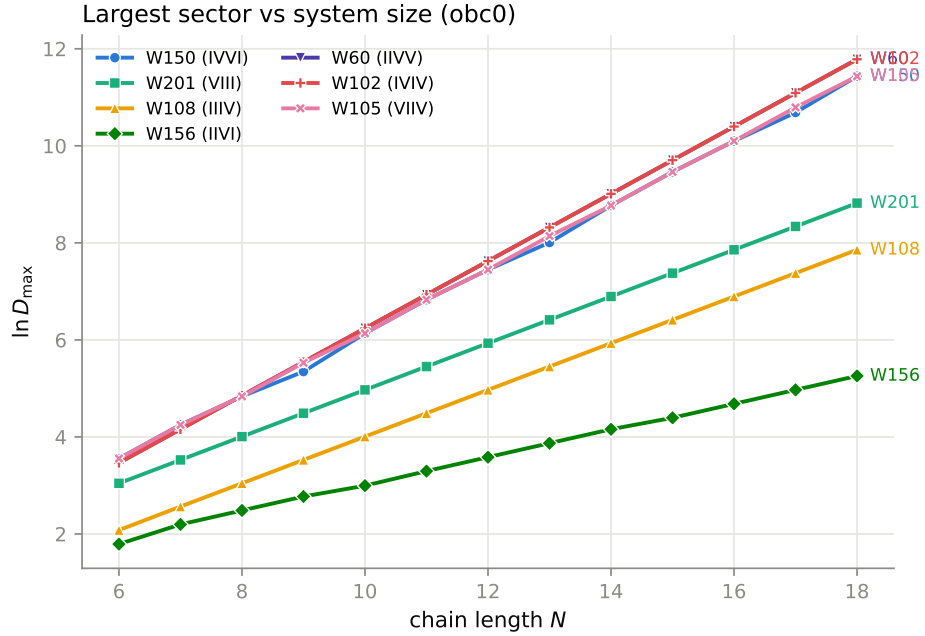


Figure 2: Largest sector $D_{\max}$ vs N (obc0), log scale. Slopes give the exponential base: φ (golden ratio) for 201, $4^{1/5} \approx 1.32$ for 156/198, 2 for the domain-wall family 60/102.

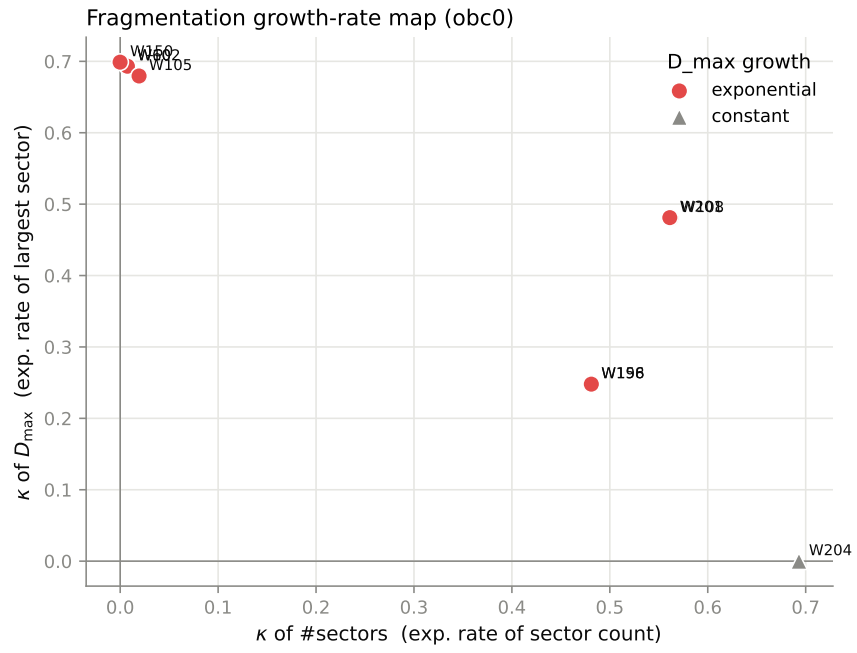


Figure 3: Growth-rate map (obc0): exponential rate of the sector count ($\kappa_{\#}$) vs. of the largest sector (κ_D), one point per rule.